

Kiranmayee Lokam

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Summary

Data Engineer with 3 years of experience building production data pipelines and ML infrastructure across Spark and Kafka streaming, Snowflake and dbt analytics engineering, production MLOps, and RAG and LLM systems. IEEE-published researcher with a track record of delivering pipelines processing 10M+ records daily at 99%+ reliability.

Experience

Associate Data Engineer

2021 – 2023

Cognizant Technology Solutions

- Built and owned 12+ production data pipelines (PySpark/Python) for 3 enterprise clients, processing 10M+ records daily.
- Reduced pipeline latency by 25% through partition tuning and optimized Spark transformations.
- Maintained 99%+ pipeline reliability across mission-critical data workflows.
- Recognized with the Sapphire Award (Q4 2023) and Employee of the Month (Q2 2023) for operational excellence.

AI Engineer / Graduate Research Assistant

2025 – 2026

California State University, Northridge

- Authored IEEE-published research on vehicle energy prediction achieving $R^2 = 0.94$ on 37,000+ OBD-II records.
- Built a production RAG pipeline with RAGAS evaluation measuring faithfulness, relevance, context precision, and recall.
- Fine-tuned Llama 3 with 4-bit QLoRA, cutting VRAM from 80GB to 8GB and improving ROUGE-L by 178%.
- Mentored NSF-REU students on ML systems and reproducible research practices.

Education

Master of Science in Computer Science

2024 – 2026

California State University, Northridge

Bachelor of Technology in Computer Science & Engineering (Big Data & Analytics)

2017 – 2021

Koneru Lakshmaiah Education Foundation

Skills

Data Engineering: PySpark, Kafka, Airflow, dbt, Snowflake, SQL, Hive, Medallion Architecture, Data Quality

AI / ML: PyTorch, CatBoost, XGBoost, Scikit-learn, SHAP, MLflow, Weights & Biases

AI Systems: LangChain, LangGraph, ChromaDB, Qdrant, HuggingFace PEFT, RAGAS, Llama 3, Prompt Engineering

Cloud & DevOps: AWS (S3, EC2, Lambda, SageMaker), Docker, Kubernetes, GitHub Actions, CI/CD, Grafana

Backend & APIs: FastAPI, REST APIs, PostgreSQL, MongoDB, Redis, TypeScript, JavaScript

Languages: Python, SQL, Java, Scala, TypeScript, JavaScript

Projects

AC Advisor — IEEE Published Research

CatBoost · SHAP · Streamlit · FastAPI · AWS

- Benchmarked 6 ML/DL models on 37,000+ OBD-II telemetry records; selected CatBoost achieving $R^2 = 0.94$, MAPE <10%, and 4.2ms inference latency.
- Deployed live Streamlit scenario analysis app on AWS with SHAP explainability; published in IEEE Xplore (2026).

Real-Time Streaming ML Pipeline

Kafka · PySpark · CatBoost · FastAPI · Docker · Prometheus · Grafana

- Architected Kafka (16 partitions) to PySpark Structured Streaming to CatBoost to FastAPI serving pipeline; processed 10M+ events/day at 4.2ms P50 latency with 99.97% uptime.
- Implemented stateful windowing, watermarking for late data, and horizontal Spark scaling with full Prometheus/Grafana observability.

Enterprise Analytics Platform

dbt · Snowflake · Airflow · SQL · Tableau · Python

- Built Bronze to Silver to Gold to Customer 360 medallion warehouse with 50+ dbt data quality tests; delivered 99.5% freshness SLA and under 2-minute query latency.
- Orchestrated Airflow DAG with Slack alerting on SLA breach; reduced analyst query turnaround from 3 days to under 5 minutes.

Production RAG Pipeline & LLM Systems

LangChain · ChromaDB · Qdrant · HuggingFace PEFT · Llama 3 · Weights & Biases

- Built RAG pipeline with 3 chunking strategies and RAGAS evaluation harness (faithfulness, relevance, context precision/recall); integrated ChromaDB and Qdrant for pluggable vector retrieval.
- Fine-tuned Llama 3 with 4-bit QLoRA reducing VRAM from 80GB to 8GB (ROUGE-L +178%, domain accuracy +163%); built LLM Observability Dashboard detecting 12% faithfulness drift via @observe SDK.

Achievements & Certifications

- IEEE Publication (2026): AC Advisor – Comparative Evaluation of ML and State Space Models for Automobile A/C Energy Prediction.
- Cognizant Sapphire Award & Employee of the Month – Recognized for operational excellence and engineering contributions.
- Salesforce Certified Administrator | Salesforce Certified AI Associate | Salesforce Certified Platform Developer I
- Automation Anywhere Certified Advanced RPA Professional | Prompt Engineering (OpenAI)